LAB 2 REPORT

Agile Backlog Creation & Sprint Simulation in Jira

Student Name	Harshith k k
Project	Warehouse Inventory & Pallet Location Tracker
Domain	Smart Cities, Transport & Logistics
Users	Warehouse Operator, Logistics Supervisor

Objective

- To convert functional requirements into Agile backlog items.
- Organize the requirements into an Epic and User Stories.
- Prioritize and estimate the User Stories.
- Simulate a Scrum sprint in Jira.
- Analyze sprint progress using the Burndown Chart.

2. Epic

Epic Name	Pallet Placement, Load Validation & Location Tracking
Description	Enable warehouse operators and logistics supervisors to place, validate, track, and locate pallets across 3D shelf positions, enforcing rack maximum load limits and capturing all stock movements via barcode/RFID scanning.

3. User Stories, Priority & Story Points

The User Stories follow the format: As a [user role], I want [goal], so that [benefit].

Issue	Story	User	User Story	Priority	Points
WPT-2	Validate Pallet Placement Against Load Capacity	Warehouse Operator	As a warehouse operator, I want the system to validate a proposed pallet placement against the rack's maximum load capacity before confirming storage, so that racks are never overloaded.	High	5
WPT-3	Assign and Record 3D Shelf Coordinates	Warehouse Operator	As a warehouse operator, I want to assign and record a pallet's 3D shelf position (aisle, rack, level) at the time of storage, so that its exact location is always known.	High	5
WPT-4	Log Stock Movements via Barcode/RFID Scanning	Warehouse Operator	As a warehouse operator, I want every inbound, outbound, and relocation movement to be logged automatically via barcode/RFID scanning, so that inventory records stay accurate without manual data entry.	High	5
WPT-5	Locate Pallet by SKU	Logistics Supervisor	As a logistics supervisor, I want to search for a SKU and instantly see the coordinates of every pallet holding it, so that I can plan picks and audits efficiently.	High	8

WPT-6	Flag Rack Overload and Discrepancy Alerts	Logistics Supervisor	As a logistics supervisor, I want to receive an alert whenever a placement attempt or a scan discrepancy indicates a rack is at or near its rated load limit, so that I can intervene before a structural or safety issue occurs.	Medium	3

Total Story Points: 26

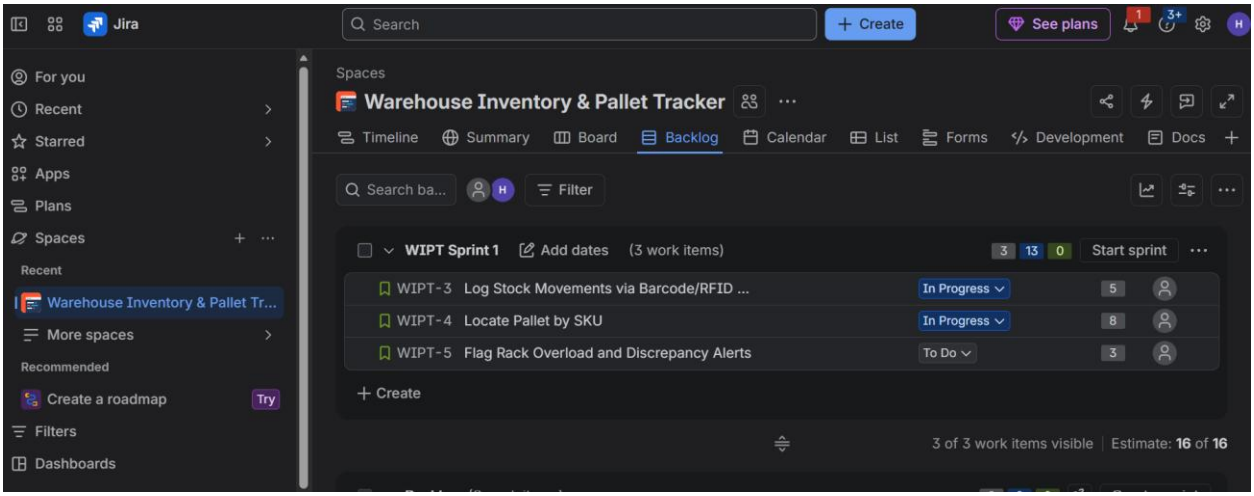
Traceability: FR-001 → WPT-2; FR-002 → WPT-3; FR-003 → WPT-4; FR-004 → WPT-5; FR-005 → WPT-6.

3.1 Non-Functional Requirements Considered

ID	Type	Description
NFR-001	Performance & Security	The warehouse item lookup service must locate any SKU's pallet coordinates across 100,000 bins in under 100 ms, under simulated peak load and standard security controls.
NFR-002	Reliability & Availability	The pallet tracking system must remain available 99.5% of the time during warehouse operating hours, with barcode/RFID scan events queued and retried on transient network loss so no movement is lost.

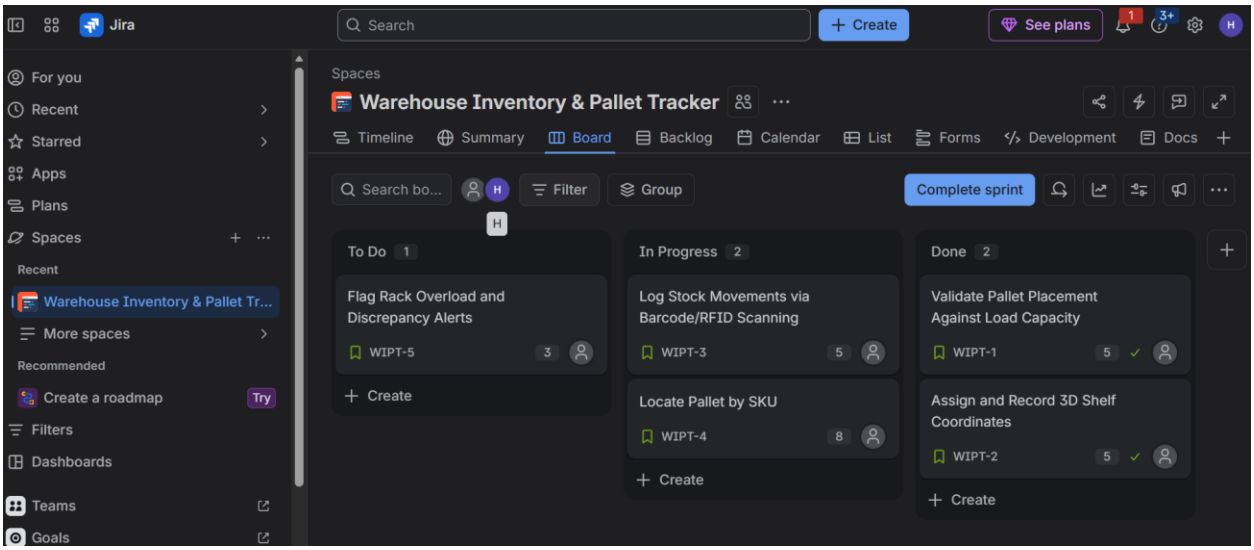
4. Jira Backlog Evidence

The Jira backlog should show the 5 work items above added to "Sprint 1" with their assigned story points, for a total sprint estimate of 26 story points.



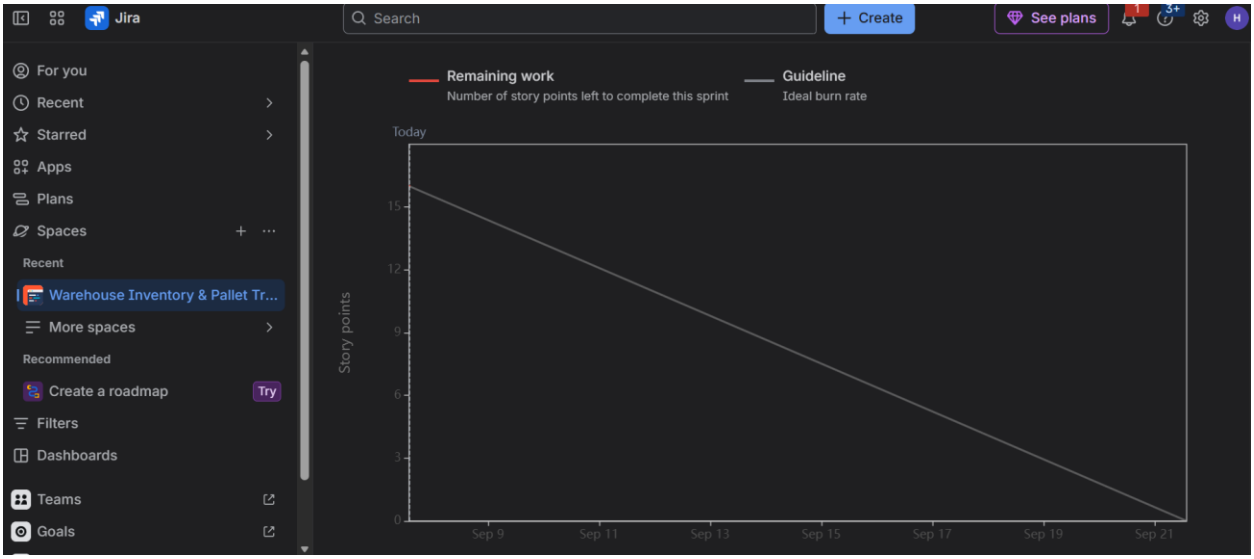
5. Sprint Simulation

Sprint 1 was created as a one-week sprint. The selected user stories were added to the sprint and simulated through the workflow: To Do → In Progress → In Review → Done.



6. Burndown Chart

The Burndown Chart was generated from Jira Reports using Story Points as the estimation statistic. The chart shows the guideline and the remaining work across the sprint timeline.



7. Reflection Questions

1. Did your estimations reflect the actual effort?

The story points were useful for estimating the relative complexity of each warehouse feature — for example, load-capacity validation and SKU lookup were sized higher than coordinate logging because they involve more edge cases (overload handling, sub-100ms lookup across 100,000 bins). Since this was a simulated sprint, actual development effort was not measured over a real implementation period, but the estimates gave a reasonable comparison of story size and complexity.

2. Was your backlog well-prioritized?

Yes. Placement validation, coordinate assignment, movement logging, and SKU-based pallet lookup were assigned High priority because they are essential to the core tracking workflow and to warehouse safety. The overload/discrepancy alert story was assigned Medium priority because it builds on top of the core placement and scanning features.

3. How did your simulated sprint align with your plan?

The sprint followed the planned workflow. The five selected stories were placed in Sprint 1 and progressed through the board. The total planned estimate for the sprint was 26 story points.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart provides a visual view of remaining work compared with the guideline. It demonstrates how completed stories (such as coordinate assignment and movement logging) reduced remaining story points over the sprint, and helps assess whether the planned workload — including the heavier SKU-lookup story — was on track within the sprint duration.